Review: *The worker is the constant-branch slice of the weld*

(MacBeth, 2026-10-18, email UID 412; rework of my UID 405 MUSTs M1–M3)

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Reviewed:	MacBeth WIP `585de70` (8 pp.; reachable from `scotmacbeth/work-in-progress` HEAD, attached PDF byte-identical to `papers/2026-10-18-weld-worker-bicomodule.pdf`)
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Check script:	`scripts/day245/macbeth412/check_weld_worker.py` (+`.log`), <1 s

Notation as yours: $\mathsf{St}_X = Xy^X$, $n = |S|$, $m = |T|$, $S, T \neq \emptyset$. A position of $B = \mathsf{St}_S \triangleleft \mathsf{St}_T$ is (s, g) with $g\colon S \to T$; a direction is (a, t'), $a \in S$, $t' \in T$ (a morphism $g(a) \to t'$ in St_T).

Verdict

- **Your hardest question: yes, B' is coaction-closed.** The reason is sharper than "λ, ρ fix g", and it shows that constancy plays no role (below). The explicit coactions are

$$\lambda(s,g) = \big(s,\ a \mapsto (a,g)\big), \quad \lambda^\sharp\big(a,(a',t')\big) = (a',t'); \qquad \rho(s,g) = \big((s,g),\ (a,t') \mapsto t'\big), \quad \rho^\sharp\big((a,t'),t''\big) = (a,t'').$$

  So λ fixes g but *moves* s to every $a \in S$, and ρ fixes the whole position. A set of positions is closed iff it is S-saturated, i.e. of the form $S \times G$. Since $B' = S \times \mathrm{const}(T)$, it is closed. **checked-sober** (by hand, and by brute force at all nine shapes $n, m \in \{1,2,3\}$; all five bicomodule laws hold on B and on B').

- **But closure does not single out B'.** Every $S \times G$, $G \subseteq T^S$, is a sub-bicomodule, the non-constant part $S \times \mathrm{NC}$ included. Each slice $B_g = S \times \{g\}$ is a minimal closed set, the structure maps above do not depend on g, and so $B \cong T^S \cdot B_{g_0}$ as bicomodules. Any injection $T \hookrightarrow T^S$ gives a bicomodule iso from A onto its image (checked with m non-constant maps). const is canonical for a different reason: it is the *unique natural* map $T \to T^S$ (Yoneda: $\mathrm{Nat}(\mathrm{Hom}(1,-), \mathrm{Hom}(S,-)) \cong \mathrm{Hom}(S,1) = 1$). Say so (S1).

- **Clause (2), Thm 5: correct as a statement about $\bar{F}_A \subseteq \bar{F}_B$**, and the verdict sentence is wrong at $(1,2)$, $(1,3)$ (M2).

- **Clause (3), Thm 6: the counts and the refutation are correct.** However, $e(s,g) = (s, \mathrm{const}_{g(s)})$ is *not* a bicomodule map. It fails λ-equivariance at every non-constant g, i.e. on exactly your interesting locus (M1). The repair is cheap and keeps every number.

- **MUST: 2. SHOULD: 3. NIT: 2.** M1–M3 of UID 405: all three addressed. M3 is fully landed, M1 is landed apart from the verdict sentence (new M2), and M2 is landed apart from the retract's type (new M1).

Grades

Claim	Grade
Lemma 2 (unit-composite $d \triangleleft c$ is a (c,d)-bicomodule)	checked-sober (standard; all 5 laws computed)
Lemma 3 (B' sub-bicomodule)	**checked-sober**; proof text needs S3
Prop. 1 ($\sigma\colon A \cong B'$)	as written **sketched** (circular, S2). With A's cobase-change structure: **checked-sober** (proof below)
Thm 5 ($\bar{F}_A = T \cdot V^S \subseteq T^S \cdot V^S$; $=$ iff $n = 1 \vee m = 1$)	**checked-sober**
Clause-(2) verdict sentence ("true when $\lvert S\rvert = 1$")	**false** at $(1,2), (1,3)$ under your own reading (M2)
Thm 6 (idempotent, image, fibre $T^{S\setminus\{s\}}$; $3 \neq 6 \neq 9$)	**checked-sober** as a map of position sets
"Worker arises from the weld by the retract e"	**false** at bicomodule level for $n, m \geq 2$ (M1); true for r_{s_0}
Thm 7 (unified)	**sketched** until M1, M2; checked-sober after
Remark 1, Example 1 numbers $(16, 64, 128;\ 8, 16;\ 4;\ 6)$	computed, all agree

Findings

MUST

M1. §6 opening, Thm 6, Thm 7: e is not a bicomodule retraction. By hand:

$$\lambda(e(s,g)) = \big(s,\ a \mapsto (a, \mathrm{const}_{g(s)})\big), \qquad (\mathsf{St}_S \triangleleft e)(\lambda(s,g)) = \big(s,\ a \mapsto e(a,g) = (a, \mathrm{const}_{g(a)})\big).$$

These differ as soon as $g(a) \neq g(s)$ for some a, i.e. for every non-constant g. The script finds the first failure at $(2,2)$, position $(0,(a,b))$, and fails likewise at $(2,3),(3,2),(3,3)$. e is ρ-equivariant. "Evaluate at the *current* input" is what breaks it, because λ changes the current input. Also, "the retract $T^S \twoheadrightarrow \mathrm{const}(T)$" is ill-typed: e needs s, so it is a map $S \times T^S \to S \times T^S$, not a map $T^S \to T$.
Fix. Either (a) say explicitly that e is a retraction of position sets only and drop "the worker arises from the weld by" it. Or (b) use $r_{s_0}(s,g) = (s, \mathrm{const}_{g(s_0)})$, $r^\sharp = \mathrm{id}$, for a chosen base input s_0. This one *is* a bicomodule retraction $B \twoheadrightarrow B'$ (checked, all shapes). It forgets $g|_{S\setminus\{s_0\}}$, with fibre m^{n-1}, so every number in (ii)–(iii) survives with $s \to s_0$. It is honestly non-canonical: natural maps $(-)^S \Rightarrow \mathrm{id}$ correspond to elements of S (Yoneda), so there is *no* choice-free bicomodule retraction when $n \geq 2$. On $\bar{F}$, which you already evaluate at a fixed St_S-object, e and r_{s_0} agree. This is probably what you meant.

M2. p. 6, "Verdict for clause (2)", and the email's "worker = content iff $|S| = 1 \vee |T| = 1$". Your Thm 5 makes the content factor the power $V \mapsto V^S$. Then $\bar{F}_A = T \cdot V^S$ equals the content iff $m = 1$. At $(1,2)$ we have $A = B$ on the nose, but $\bar{F}_A = 2V \neq V$. So the "true when $|S| = 1$" half is false as literally stated. What Thm 5 proves is *worker = whole weld* ($\bar{F}_A = \bar{F}_B$, and $A = B$) iff $n = 1 \vee m = 1$. Under the literal reading, clause (2) (*worker = content factor*) holds iff $m = 1$, and fails at $(1,2)$, $(1,3)$ as well as on $n, m \geq 2$. Pick one reading and state it. I share the blame here: my UID 405 M1 wrote "$B = A$ on the nose, yet Theorem 6 says the worker is not the content", which runs the two readings together.

SHOULD

S1. Lemma 3 and Def. 3 framing. The proof never uses constancy, so state the general fact: the sub-bicomodules supported on position subsets are exactly $S \times G$, and $B \cong \coprod_{g \in T^S} B_g$ with all B_g isomorphic. Then "B' is canonical" rests on Yoneda (diagonal = unique natural $T \to T^S$), not on closure. Positive control in the script: $S \times \mathrm{NC}$ is closed at every shape.

S2. Prop. 1 is circular as written, and Remark 2's caveat can go. Thm 1 *defines* A's structure by transport along σ, which makes σ an iso trivially. The non-trivial statement uses the cobase-change structure of Remark 2, which takes three lines. Take the cofunctors $\pi_S(s,t) = s$, $\pi_S^\sharp(a) = (a,t)$ and $\pi_T(s,t) = t$, $\pi_T^\sharp(t') = (s,t')$. Then $\lambda_A = (\pi_S \triangleleft A)\delta_A$ and $\rho_A = (A \triangleleft \pi_T)\delta_A$ give $\lambda_A(s,t) = (s, a \mapsto (a,t))$, $\lambda_A^\sharp(a,(a',t')) = (a',t')$, $\rho_A(s,t) = ((s,t),(a,t') \mapsto t')$ and $\rho_A^\sharp((a,t'),t'') = (a,t'')$. Apply σ and you get the displayed λ, ρ of B' with $g = \mathrm{const}_t$, since $\lambda_B(s, \mathrm{const}_t) = (s, a \mapsto \sigma(a,t))$. So σ is a bicomodule iso from the *intrinsic* structure. All five laws for A and both equivariances of σ are computed at all nine shapes, and $\bar{F}_A(V) = T \cdot V^S$ is recomputed from A's own structure, not via σ.

S3. Lemma 3 proof text. (a) The ρ bullet: in $B \triangleleft \mathsf{St}_T$ the copy of B is *outer*. ρ keeps (s,g) entirely and labels direction (a,t') by t'. (b) "$\delta_T(t,\psi)$": positions of St_T are bare $t \in T$, and there is no ψ. Likewise $\delta_S(s,\varphi)$. (c) The λ bullet should say that λ moves s, which is why the closure needs S-saturation and not just "g fixed". My negative control NEG1 is exactly this: $\{(s_0, \mathrm{const}_t)\}$ is not closed once $n \geq 2$.

NIT

N1. §6 end, "It is nonempty — the retract actually loses something". Fibres always have $m^{n-1} \geq 1$ points. You mean the fibre is *non-trivial* (> 1 point). Same for "empty discard" in Thm 7.

N2. Thm 6(iii) is true but weaker than it reads: the fibre omits *every* g with $g(s) \neq t$, constant or not. Drop it, since (i) and (ii) already refute the claim.

Controls (the test separates)

- **NEG1** (non-saturated subset $\{(s_0, \mathrm{const}_t)\}$): closure FAILS at every shape with $n \geq 2$ and passes iff $n = 1$, as predicted.
- **NEG2** (a *valid* twisted left coaction, $S = T = \mathbb{Z}/n$, $\lambda_{\mathrm{tw}}(s, g) = (s, a \mapsto (a, g + (a - s)w))$, $w = \delta_0$ non-constant): all five laws PASS and the closure of B' FAILS at $(2, 2)$ and $(3, 3)$. So the closure test is not vacuous for genuine bicomodules, and it is the specific form of λ that makes B' closed.
- **NEG3** (invalid bump of g in each λ-slot): counit and coassociativity FAIL, and closure FAILS, at every $n, m \geq 2$.
- The retract test separates too: e fails λ-equivariance and r_{s_0} passes, at the same shapes.

Appendix: script output (abridged, shape $(2, 3)$; full log in repo)

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B laws: lam counit, lam coassoc, rho counit, rho coassoc, compat: all True
lam(s,g)=(s,a->(a,g)) True; lam#(a,(a',t))=(a',t) True; rho(s,g)=((s,g),(a,t)->t) True; rho#((a,t),t'')=(a,t'') True
(i) B' = S x const(T) closed under lam, rho: True
every slice S x {g} is a minimal closed set (so closed subsets = S x G, any G <= T^S): True
positive control: NC = S x (non-constant) also closed: True
NEG1 non-S-saturated {(s0,const_t)} closed: False
A (cobase change of regular A along projections) laws: all True
Prop 1: sigma(s,t)=(s,const_t), sigma#=id is a bicomodule map A -> B: True; injective True; image = B' True
sigma_G onto S x G, G = 3 NON-constant maps, also a bicomodule iso onto image: True
V=2: F_B=36 (=T^S V^S), F_B'=12, F_A(own)=12 (=T V^S), content V^S=4
M2: e idempotent True; image = B' True; fibre sizes [3] = T^(S-1) True; |NC|=6 |T^S|=9
e is rho-equivariant: True; e is lam-equivariant: False  <-- FAILS at (0, ((0, 'a'), (1, 'b')))
r_s0(s,g)=(s,const_g(s0)): bicomodule map True, retraction onto B' True, fibre 3
shapes (1,1),(1,2),(2,1),(1,3),(3,1),(2,2),(2,3),(3,2),(3,3): ALL ASSERTIONS PASSED
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